
Feature Learning in Supervised AI Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 In this research paper we will learn and investigate how various supervised learning
2 models learn features. We will do this through four tasks. In task 1 we will see
3 how strong inductive bias is in models and how their architecture can effect how
4 they learn features (i.e. whether they learn shape or texture or some other attribute).
5 In task 2 we will observe how this effects Domain Generalization. The idea being
6 that we want the model to learn identifying features about the objects (like their
7 shape) instead of Domain Specific features (like if the dataset had grass behind
8 every cat and the model associated the grass with the cat). If the model can do this,
9 it should perform well even on Domain Shifts. We extend this idea in task 3 with
10 Domain Generalization and completely unknown target domains. Finally, in task 4
11 we tackle the open set problem, to see if we can identify casual features even on
12 unknown classes, and separate them as they come.

13 1 Introduction

14 We will do various experiments in this research paper to study how different AI models learn features,
15 and the consequence of their architectures. This is done in four parts.

16 First, we will see the study the influence of Inductive Biases and Feature Representations, our goal
17 here is to see the influence of model architecture. By comparing the metrics of four well known and
18 standard models (ResNet-50, ViT-B, CLIP). We will first perform a baseline metric on all of them,
19 and then perform 5 different experiments. Colour Bias: We perform two different types, grey scale,
20 and hue shift, and see how well the models perform. Shape vs Texture (cue-conflict): We apply the
21 texture of one object to another, to see if models are using texture or shape for classification and
22 which is more important. Translation: We shift the images, left, right, up, or down and see if the
23 model can still classify. Lastly, we shuffle the patches of the images to see how the global structure
24 effects the models.

25 Our points of interest are, what cue-conflict and colour reveal about reliance on shape, texture, and
26 colour, and what coverage can tell us about shape bias. How the models respond when spatial structure
27 is altered and what that tells us about the importance of locality, global organisation, and positional
28 sensitivity. To what extent prediction-level changes agree with changes in feature representations,
29 and which observed difference can reasonably be attributed to architecture as opposed to pretraining
30 data, supervision, augmentation or model capacity.

31 2 Experimental Setup/Methods

32 2.1 Inductive Biases and Feature Representations

33 Models:

34 • torchvision ResNet-50 with ResNet50_Weights.IMAGENET1K_V2;

- torchvision ViT-B/16 with ViT_B_16_Weights.IMAGENET1K_V1;
- OpenCLIP ViT-B-32 with pretrained='openai'.

Dataset: STL-10

Configuration:

- Seed: 6304
- Optimizer: AdamW
- Learning Rate: 10^{-3}
- Weight Decay: 10^{-4}

For the Shape vs Texture we used AdaIN implementation, which follows Huang & Belongie (2017). the encoder/decoder architecture and the pretrained weights come from the public reimplementation naoto0804/pytorch-AdaIN, weights mirrored at BhupatiNadar/adain-style-transfer-models on HuggingFace.

3 Results

3.1 Inductive Biases and Feature Representations

We first take some basic measurements as a baseline, we freeze the three classifier heads and measure top-1 accuracy, macro-F1, and mean maximum confidence. We also do this for zero shot CLIP, we measure confidence from the softmax over scaled class similarities.

Table 1: Clean Baseline - Shows all models performing very well on the dataset.

System	Top-1 (%)	Macro-F1 (%)	Mean max conf.
ResNet-50 + linear head	96.4	96.39	0.91
ViT-B/16 + linear head	97.2	97.21	0.97
CLIP ViT-B-32 + linear head	96.6	96.60	0.27
CLIP ViT-B-32 zero-shot	97.0	97.01	0.94

All the models perform well as a baseline, this maybe a limitation of the STL-10 dataset, as its distinct classes would be easier to classify. This can be a limitation as differences in experiments would more visibly show a decrease rather than an increase. It should also be noted that the CLIP Max Mean is low because it uses Cosine similarity in training, and the L-2 norm forces the magnitude to be 1. So when attached with a linear head and no compensation, the value comes off as low. It does not mean it is lower in confidence than other models, just that rows can't be compared the magnitude lower, not because it is less confident than others

3.1.1 Color Bias

We did two experiments with colour, the first we changed the images to greyscale measured the same metrics and got the following results:

Table 2: Greyscale: Minimal change in Top-1 Accuracy

System	Top-1	Macro-F1	Δ Top-1	Consistency	Conf.
ResNet-50 + linear head	94.6	94.61	-1.8	95.8	0.91
ViT-B/16 + linear head	95.4	95.41	-1.8	98.0	0.96
CLIP ViT-B-32 + linear head	93.8	93.78	-2.8	93.0	0.25
CLIP ViT-B-32 zero-shot	93.0	92.97	-4.0	94.0	0.93

We can see that making the image greyscale results in a slight decrease in accuracy throughout. Given that we have removed information from the image this is to be expected. A more interesting

64 observation is that there was no significant difference between Convolution and Vision Transformers,
65 though consistency is slightly higher for the Vision Transformer. CLIP had the most sever impact.
66 Both of the readings are lower than either ResNet or ViT-B, and further zero-shot is the lowest.
67 One plausible explanation for this is that the text embeddings had an influence on how accurately
68 important colour is during pre training. For example if a prompt were to be "A picture of a Brown
69 Dog." that would influenced both the visual encoder and the text encoder. It would also explain why
70 zero-shot performed even worse. This though is not conclusive, and would be hard to confirm.

71 The second experiment was with a Hue Shift of 40 deg. A limitation of this is that a hue shift will only
72 have a good effect on well saturated images. On less saturated images the change is very minimal,
73 making the result less significant.

Table 3: Hue Rotation of 40: Minimal change in accuracy

System	Top-1	Macro-F1	Δ Top-1	Consistency	Conf.
ResNet-50 + linear head	94.4	94.42	-2.0	95.2	0.90
ViT-B/16 + linear head	95.8	95.79	-1.4	98.2	0.96
CLIP ViT-B-32 + linear head	94.8	94.78	-1.8	95.8	0.26
CLIP ViT-B-32 zero-shot	95.2	95.18	-1.8	95.6	0.93

74 Again we see a minimal change in accuracy, this time they are much closer to each other. No claims
75 can be made as the difference is small enough that it can be considered noise. Again though, we will
76 find that ViT is just slightly more consistent than other models. It is also of note that unlike greyscale,
77 CLIP is much more in line with other models. One explanation for this is that in greyscale, we totally
78 lost colour information, however with a rotation, the relationship between the colour of the objects is
79 still the same, it has just been rotated. Hence, the loss is less with CLIP now.

80 3.1.2 Shape vs. Texture

81 For this comparison alpha strength of 1 was required, as lower values did not transfer texture well
82 enough. We get the following results:

Table 4: Shape vs Texture: Convolutional models are not shape biased.

System	Balanced ($n = 220$)					All accepted ($n = 286$)	
	N_{sh}	N_{tex}	N_{oth}	Shape bias	Coverage	Shape bias	Coverage
ResNet-50 + linear head	54	52	114	50.94	48.18	57.78	47.20
ViT-B/16 + linear head	99	14	107	87.61	51.36	89.94	55.59
CLIP ViT-B-32 + linear head	116	12	92	90.62	58.18	91.72	59.09
CLIP ViT-B-32 zero-shot	117	7	96	94.35	56.36	95.81	58.39

83 Out of a total of 400 candidate images only 286 were of good enough quality, which were further
84 reduced to 220 to make each class balanced. As expected, the Vision Transformers have a considerable
85 shape bias, while the convolutional model does not. Though one point of note is that ResNet still
86 managed to score a 50.94 on the Shape Bias. This is considerably higher than the original Geirhos et
87 al. (2019) paper. The reason for this can be mostly likely presumed to be the biggest limitation of
88 this experiment.

89 That is that the images were filtered out manually. This introduced significant human bias in the
90 process making the data unreliable. The human eye is considered to be heavily biased towards shape,
91 I personally noticed as well that during the filtering it was much easier for me to tell when the shape
92 data was destroyed in comparison to when texture data was to sufficiently applied. It is reasonable to
93 then conclude the data had a bias towards higher quality shape images than texture images.

94 Another notable observation is that for ViT and ResNet both, there most quantitative category is
95 neither shape nor texture, and CLIP barely has more shape classified images. Making the coverage
96 very low, and limiting our ability to draw further conclusions from this data. This can again be

contributed to the low quality of the data. The AdaIN implementation resulted in images with quite high distortion with our particular dataset, hampering the ability of the models to classify.

3.1.3 Patch Structure

We divided the images into 4x4 patches, and shuffled the patches. We feed the shuffled images back in the model to measure the metrics:

Table 5: Patch Shuffling: Transformers suffer the most

System	Top-1 (%)	Macro-F1 (%)	Δ Top-1	Consistency (%)	Mean max conf.
ResNet-50 + linear head	89.0	89.06	-7.4	90.4	0.87
ViT-B/16 + linear head	90.4	90.52	-6.8	92.6	0.89
CLIP ViT-B-32 + linear head	84.4	84.18	-12.2	84.6	0.20
CLIP ViT-B-32 zero-shot	84.4	84.20	-12.6	84.2	0.82

When we shuffle the images we destroy the global structure of the image, i.e. the shape. For Vision Transformers this is quite bad as they use the self attention mechanism and look at a global picture. Understandably, they suffer the most in accuracy. Convolutional networks however only look at the local area, making them much more resilient to this, further texture is preserved more, as each patch can act like a texture patch and we saw in table4, they are much more biased towards texture.

The ViT-B and ResNet model are close in accuracy. This is a bit unexpected because we discussed ResNet should perform noticeably better here. The likely culprit is that we used zero padding, max pooling, stride 2, and we averaged the final 7x7 map which all contribute to either a lack of local information or hampers its readability for the models.

One final thing of note is that CLIP performs noticeably worse than ViT, even though they share the visual encoder (as a Vision Transformer), the fact that both zero-shot and linear head land at the same number implies that its not the involvement of the text encoder in the prediction process. This suggest that contrastive learning (with the text encoder) during pre training or some other stage had some factor that made them more dependent factor.¹

3.1.4 Representation Analysis

We have discussed most experiments now, but those mainly measured when the predictions changed and hence also changed the accuracy. But how the model sees and interprets the feature can change even if the prediction didn't. We measure the Cosine Similarity and t-SNE maps where relevant.

Table 6: Cosine similarities - Expected Results

Intervention	ResNet-50		ViT-B/16		CLIP ViT-B-32	
	I_T	Norm.	I_T	Norm.	I_T	Norm.
Grayscale	0.789	0.719	0.737	0.711	0.900	0.743
Translation (32 px)	0.934	0.912	0.943	0.938	0.970	0.923
Patch shuffle	0.718	0.626	0.649	0.615	0.728	0.299
Cue conflict vs. content	0.250	0.004	0.223	0.146	0.651	0.101
Cue conflict vs. style	0.216	-0.041	0.072	-0.020	0.580	-0.083
Random-pair floor	0.247	0.000	0.090	0.000	0.612	0.000

In terms of grayscale and translation we can see the Norm is rather high. This is consistent with our earlier hypothesis that models don't look at colour for classification. And since in translation most features are preserved (shape, texture, colour) and the image is only moved, they are very close to the original in representation.

In Patch shuffling the values a bit lower (around 0.6), which is to be expected as the global structure is destroyed. What is more interesting is that CLIP and ViT show a large difference despite them

¹We also saw more shape bias for them in table4, but it was not enough of a difference to warrant any conclusion

both having a Vision Transformer. One plausible explanation is that since CLIP’s tokens are twice as large as ViT, they are more dependent on global structure, and the representation drifts more. The larger token size could also mean that it covers the boundaries more between patches.

The t-SNE diagrams (Figures 3.1 2 in the appendix) agree with these results.

3.2 Unsupervised Domain Adaptation

3.2.1 Source Only ERM

We first train both the model and the classifier here on the source domains using cross entropy, which results in these measurements.

Table 7: Source only ERM - the main domain gap identified

Domain	Accuracy (%)	Macro-F1 (%)
Photo (source, val)	97.30	96.85
Art Painting (source, val)	91.71	91.69
Cartoon (source, val)	95.74	96.00
Mean source	94.91	94.85
Worst source	91.71	91.69
Sketch (target)	61.59	63.09

As we can see in source only statistics, the accuracy on our source domains is very good, but when ran on a unknown domain (for us that is sketch) we notice a massive accuracy drop. compared with the mean it is a -33 % drop, this is the main figure we try to lower in these experiments.

Table 8: Target Domain by class - high amounts of variation.

Class	Sketch accuracy (%)
Dog	21.11
Elephant	94.73
Giraffe	77.95
Guitar	70.07
Horse	46.32
House	83.75
Person	61.25

when we see the by class accuracy on the target domain, dog performs worst while elephant performs best. Dog is the largest class in the sketch data, we found dog’s 333 most common errors go to elephant, 167 to giraffe. It is probable it is not confusing animals and objects at least, it is confused within animal classes.

141 3.2.2 DAN - MMD Alignment

Table 9: DA with MMD - Slight improvement over Source Only

	Source-only	DAN ($\lambda_{\text{MMD}} = 1$)
Photo val acc (%)	97.30	95.80
Art Painting val acc (%)	91.71	93.41
Cartoon val acc (%)	95.74	95.95
Mean source acc (%)	94.91	95.05
Mean source macro-F1 (%)	94.85	94.93
Sketch acc (%)	61.59	64.44
Sketch macro-F1 (%)	63.09	60.55
Δ Sketch acc	—	+2.85
Domain separability (%)	100.00	81.87

142 If we observe the general by class results for MMD, we see a slight but still meaningful improvement
 143 over the sketch accuracy. Though the sketch F1 - macro decreases by a bit.

Table 10: MMD by Class - Good by class improvement

Class	Source-only (%)	DAN (%)	Δ
Dog	21.11	47.93	+26.82
Elephant	94.73	75.54	−19.19
Giraffe	77.95	61.62	−16.33
Guitar	70.07	74.18	+4.11
Horse	46.32	57.60	+11.28
House	83.75	100.00	+16.25
Person	61.25	86.25	+25.00

144 We see all classes improve in accuracy, except for giraffe and elephant, but given that in source only
 145 most of the errors in the dog class went to giraffe and elephant, it is likely they were over represented
 146 in the original model in the first place.

147 3.2.3 DANN - Adversarial Alignment

148 We now use adversarial learning to see how well it performs. And get the following results:

Table 11: DANN - The performances worsened

	Source-only	DANN
Photo val acc (%)	97.30	95.80
Art Painting val acc (%)	91.71	92.93
Cartoon val acc (%)	95.74	93.60
Mean source acc (%)	94.91	94.11
Mean source macro-F1 (%)	94.85	94.03
Sketch acc (%)	61.59	55.97
Sketch macro-F1 (%)	63.09	58.23
Δ Sketch acc	—	−5.62
Domain separability (%)	100.00	99.31

149 Interestingly not only is the performance on the sketch pictures worse than source, but the the
 150 predictions on source only is also worse. At first this might make it look like DANN didn't work, but
 151 after further investigation we find this:

Table 12: Tracking Per Epoch - Loss was decreasing while Domain accuracy was going to 0.5

Epoch	Cls. loss	Domain loss	Domain acc.	Mean source F1 (%)
1	0.533	0.384	0.817	91.37
2	0.234	0.597	0.673	89.19
3	0.159	0.669	0.597	94.03
4	0.261	0.742	0.542	84.62
5	0.176	0.686	0.551	92.80
6	0.084	0.681	0.541	92.37
7	0.070	0.694	0.502	93.92
8	0.081	0.694	0.501	93.90

Selected epoch in bold. Chance domain accuracy is 0.5 and $\ln 2 = 0.693$.

Table 13: CDAN: Significant Improvement from before

	Source-only	CDAN
Photo val acc (%)	97.30	97.30
Art Painting val acc (%)	91.71	91.46
Cartoon val acc (%)	95.74	95.52
Mean source acc (%)	94.91	94.76
Mean source macro-F1 (%)	94.85	94.66
Sketch acc (%)	61.59	74.65
Sketch macro-F1 (%)	63.09	73.85
Δ Sketch acc	—	+13.06
Domain separability (%)	100.00	99.86

The Loss was indeed decreasing, and the Domain Accuracy was going to 0.5., what this means is that even if we fool the discriminator. that does not necessarily mean an increase in performance. Perhaps the backbone managed to fool the discriminator before the discriminator got good enough to warrant an improvement. An increase in learning later or even just running more Epochs could improve metrics to match MMD or beyond.

3.2.4 CDAN - Class-Conditional Adversarial Alignment

To improve the scores from before we use the values of the prediction as well, we do a cross product with the latent and feed that to the discriminator.

The performance on the source only metric stays the same, but the accuracy on sketch images received a significant improvement. Meaning that even soft information about the final prediction was very helpful to the discriminator. Improving its performance which improved the backbone to more aggressively compete with it. However, it was not lower domain separability that did this, as domain separability essentially remained the same. It can be speculated that domain information is important as that helps the model to classify them.

3.2.5 Common Evaluation and Alignment Diagnostic

We compare the results overall:

Table 14: Diagnostic - Separability might not be the deciding factor

Method	Source val. acc. (%)			Mean source (%)		Sketch (%)		Δ Acc.	Sep. (%)
	Photo	Art	Cartoon	Acc.	F1	Acc.	F1		
Source-only	97.30	91.71	95.74	94.91	94.85	61.59	63.09	—	100.00
DAN	95.80	93.41	95.95	95.05	94.93	64.44	60.55	+2.85	81.87
DANN	95.80	92.93	93.60	94.11	94.03	55.97	58.23	−5.62	99.31
CDAN	97.30	91.46	95.52	94.76	94.66	74.65	73.85	+13.06	99.86

Sep. = domain separability: held-out accuracy of a logistic-regression probe ($C = 1$, balanced, 70/30 split, seed 6304) trained to tell source from target from the frozen features. Chance is 50%. Target labels were used only at this stage.

168 We would assume that bringing the domains closer and making them indistinguishable would be key,
 169 as the overlapping elements between the target domain and source domains would come into play.
 170 However aside from MMD which tries to directly to do this with the Loss function, a freshly trained
 171 logistical regressor can still separate the classes. And our best performance from CDAN barely shows
 172 any change, implying it might not be the most key factor.

Table 15: Per Class Comparison: Four classes share a trend

Class	Sketch accuracy (%)				Δ vs Source-only		
	Source-only	DAN	DANN	CDAN	DAN	DANN	CDAN
Dog	21.11	47.93	42.10	64.25	+26.82	+20.99	+43.14
Elephant	94.73	75.54	35.00	80.54	−19.19	−59.73	−14.19
Giraffe	77.95	61.62	51.53	62.95	−16.33	−26.42	−15.00
Guitar	70.07	74.18	98.36	89.31	+4.11	+28.29	+19.24
Horse	46.32	57.60	48.04	71.69	+11.28	+1.72	+25.37
House	83.75	100.00	100.00	100.00	+16.25	+16.25	+16.25
Person	61.25	86.25	98.12	99.38	+25.00	+36.87	+38.13

Table 16: Two most wrong - Similar pattern

True class	Source-only	DANN	CDAN
Dog	elephant 333, giraffe 167	person 397, house 18	person 139, elephant 62
Horse	giraffe 293, elephant 111	dog 204, person 166	dog 134, person 33
Elephant	person 21, giraffe 13	person 400, dog 49	person 65, dog 58
Giraffe	elephant 99, horse 34	person 164, dog 142	dog 175, horse 56

Two most frequent wrong predictions per class, with counts.

173 Further data shows us that four classes show interesting behavior. Dog (+20.99 to +43.14) and Person
 174 (+25.00 to +38.13) improve under every method, while Elephant and Giraffe degrade under every
 175 method. It almost seems like the model is trading the improvement in two for the others, no one
 176 improved all four.

177 3.2.6 Controlled Design Study

Table 17: Control study - too high a λ collapses the model

λ_{MMD}	Mean source F1 (%)	Sketch acc. (%)	Domain separability (%)
0 (source-only)	94.85	61.59	100.00
0.1	94.73	62.31	92.17
1	94.93	64.44	81.87
10	5.07	4.07	60.03

178 A higher λ means a higher penalty for difference between the source domain feature distribution, so
 179 increasing it should mean the model more aggressively tries to match the feature distributions. And

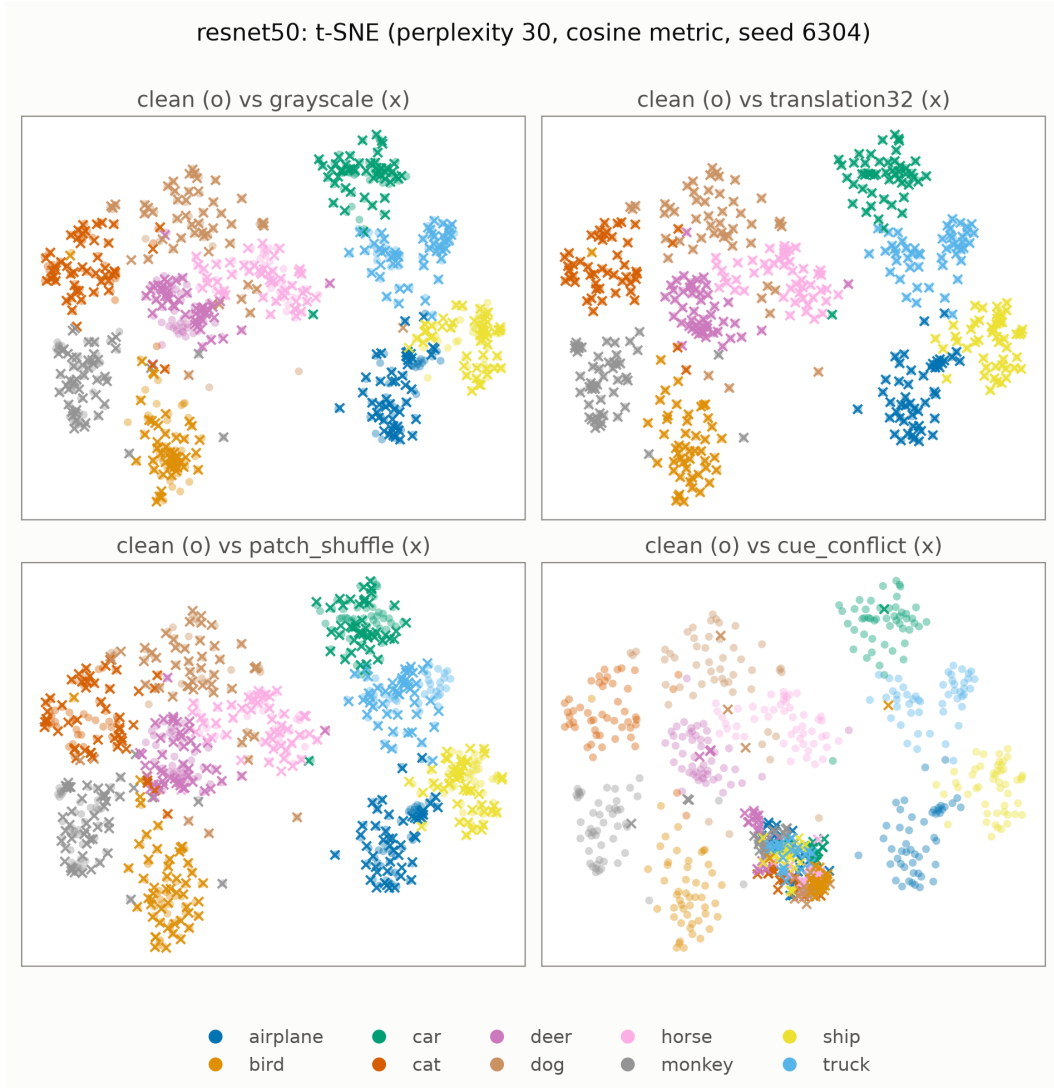


Figure 1: ResNet t-SNE: Representations overlap except in Cue Conflict

we see exactly that for the first three rows, however $\lambda = 10$ completely collapses the model. It is such a severe penalty that all the distributions get so mixed up that the model can no longer tell any of them apart. F1 Mean on source alone is enough to disqualify 10 as optimal even without consulting target domain labels.

3.3 Open-Set Recognition (OSR)

3.3.1 Vanilla Closed-Set Baseline

3.4 Cross-Task Discussion

4 Conclusions

5 Appendix

6 References

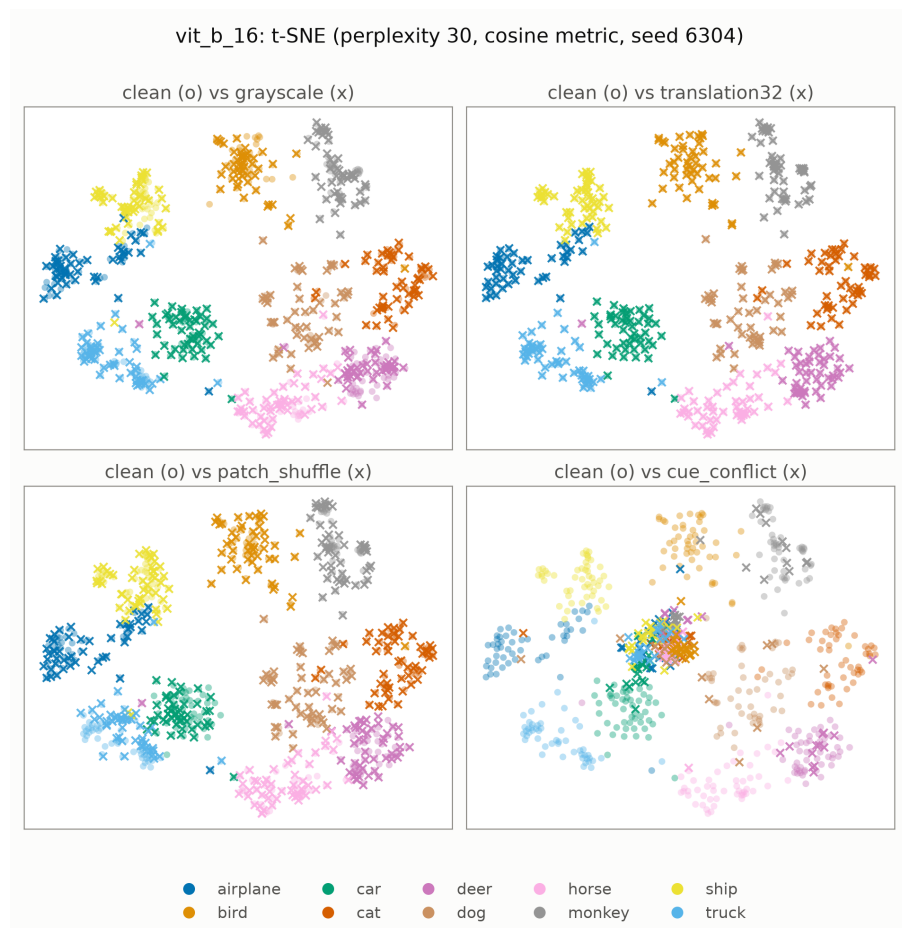


Figure 2: ViT-B t-SNE: Representations overlap except in Cue Conflict

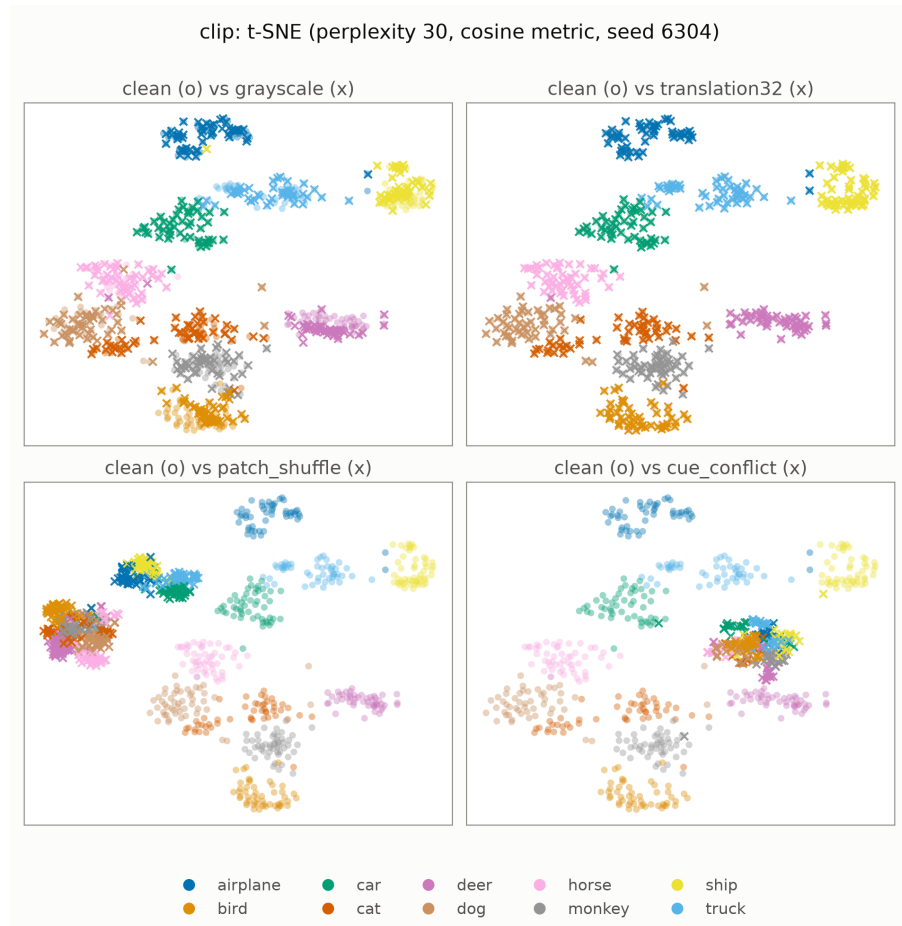


Figure 3: CLIP t-SNE: Representations overlap except in Cue Conflict and Patch Shuffle